Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Christopher Detrick	detrickc@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I want to make a free program that lets users turn vocabulary (terms and definitions) into multiple choice questions that they can use to study.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Websites such as Quizlet already attempt to address this problem, but they cost money to get full access.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

My site would let the user input several terms and definitions, and then the user would be able to take a multiple choice quiz where they have to choose the correct term based on a given definition. The site would track which questions were guessed incorrectly and would make the user answer those ones again until they get it correct.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

My program will allow the user to enter their vocabulary terms and definitions and study them using multiple-choice questions.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will need to provide as many terms and definitions as they would like, with a minimum of 4.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Each term and definition will be stored in a dataclass. All of the terms and definitions will be stored in a list of dataclasses. When the quiz starts, the definitions will be displayed in a random order, each with four buttons that show a term. There will always be one button with the correct term that matches with the definition and three buttons that have terms from other definitions, which will be randomly selected.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

When the user answers the multiple choice questions, they will receive feedback on whether they got the question correct or incorrect. If the user chose incorrectly, the term will be repeated in a later question.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Terms/Definitions, Correct Answers, Incorrect Answers

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

• Details I need: Term name, definition description, number of terms • Details I will ignore: Term length, definition length, specific contents of definition description

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Turning a list of terms/definitions into randomized multiple choice questions

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

If I end up using an LLM to help code, I will avoid shallow learning and hallucinations by having it code small portions of the larger program at a time so that I can dissect each part as I go.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal for this project is to create something useful and challenge the skills that I have learned from taking this class.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool that lets users turn vocabulary (terms and definitions) into multiple choice questions that they can use to study. What are 3 common problems students face when practicing vocabulary, and what are 3 simple features a basic tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

ChatGPT gave three common problems that students face: passive studying, the fact that creating practice questions manually is time consuming, and not properly keeping track of which terms they need more practice with. Some features it suggested were vocabulary entry, multiple-choice quiz generator, a results page that shows the user a score at the end, and saving vocabulary sets with browser local storage.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I asked for a simpler alternative to storing vocabulary sets with browser local storage. ChatGPT gave the following response: A much simpler alternative is to **keep the vocabulary only while the program is running**. Instead of saving sets for later, students enter their vocabulary, take the quiz, and see their results during the current session. When they refresh or close the page, the vocabulary can disappear. For your three basic features, I'd use:

1. **Enter vocabulary** — Add terms and definitions to a temporary list.
2. **Take a multiple-choice quiz** — Generate questions using the entered vocabulary.
3. **Show quiz results** — Display the final score and which terms were answered incorrectly.

That keeps the project focused on basic Python concepts like **lists, records/classes, functions, conditionals, and loops** rather than browser storage.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

After this conversation, I now have three simple features that I want to implement as my main goal. I now plan to focus solely on these features. The main idea of my project has not been altered, however.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck, I might use Google, documentation, an LLM, or reach out to a TA.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I would first take a look at the code and see if I can find the problem myself. If not, then I would ask the LLM to see if it could find its own mistake. If not, then I would consult a TA.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I want to build a simple web tool that lets users turn vocabulary (terms and definitions) into multiple choice questions that they can use to study. What are 3 common problems students face when practicing vocabulary, and what are 3 simple features a basic tool should have?

For your vocabulary study tool, this is a good scope for the brainstorming portion of your project worksheet.

3 common problems students face

1. **Passive studying** — Students may just reread terms and definitions instead of testing whether they can actually remember them.
2. **Creating practice questions takes time** — Turning a vocabulary list into useful multiple-choice questions manually can be tedious.
3. **Knowing what needs more practice** — Students may have trouble keeping track of which terms they consistently get wrong.

3 simple features for your tool

1. **Vocabulary entry** — Let users enter a term and its definition and build a list of vocabulary items.
2. **Multiple-choice quiz generator** — Show a term with four possible definitions: one correct definition and three definitions taken from other vocabulary items as distractors.
3. **Score/results page** — Keep track of correct and incorrect answers and show the user's score at the end.

These features are realistic for a **CS1 Drafter project** because you can represent the vocabulary with basic Python data, generate questions from that data, and track a score without needing a backend or database. If you want persistence, saving vocabulary sets with browser local storage could be an optional extension rather than part of the minimum version.

saving vocabulary sets with browser local storage is too complicated for an introductory programming project. can you give a simpler alternative?

A much simpler alternative is to **keep the vocabulary only while the program is running**.

Instead of saving sets for later, students enter their vocabulary, take the quiz, and see their results during the current session. When they refresh or close the page, the vocabulary can disappear.

For your three basic features, I'd use:

1. **Enter vocabulary** — Add terms and definitions to a temporary list.
2. **Take a multiple-choice quiz** — Generate questions using the entered vocabulary.
3. **Show quiz results** — Display the final score and which terms were answered

incorrectly.

That keeps the project focused on basic Python concepts like **lists, records/classes, functions, conditionals, and loops** rather than browser storage.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.